



PHOT 301: Quantum Photonics

LECTURE 00

Michaël Barbier, Fall semester (2026-2027)

COURSE INFORMATION

Instructor

Dr. Michaël Barbier

e-mail: michaelbarbier@iyte.edu.tr

Office: Z-5/B

Hours: Friday 9:00-17:00 (via
appointment)

Teaching Assistants

Yağız Oyun

e-mail: yagizoyun@iyte.edu.tr

Office: Z-9/B

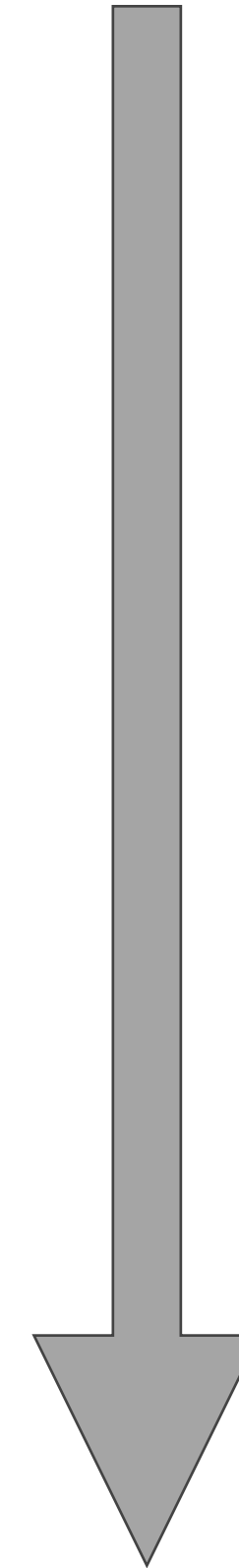
hours: TBA

Course Schedule

Thursday	13:30 – 15:15	Building F, lecture room D3
Thursday	15:30 – 17:15	Building F, lecture room D3

CONTENTS OF THE COURSE

- Wave function & Schrödinger's equation
- (Mathematical) Dirac notation
- Bound states and scattering
- Hydrogen atom, Multi-electron atoms
- Molecules and materials
- Electromagnetic-field, absorption/emission



COURSE MATERIALS

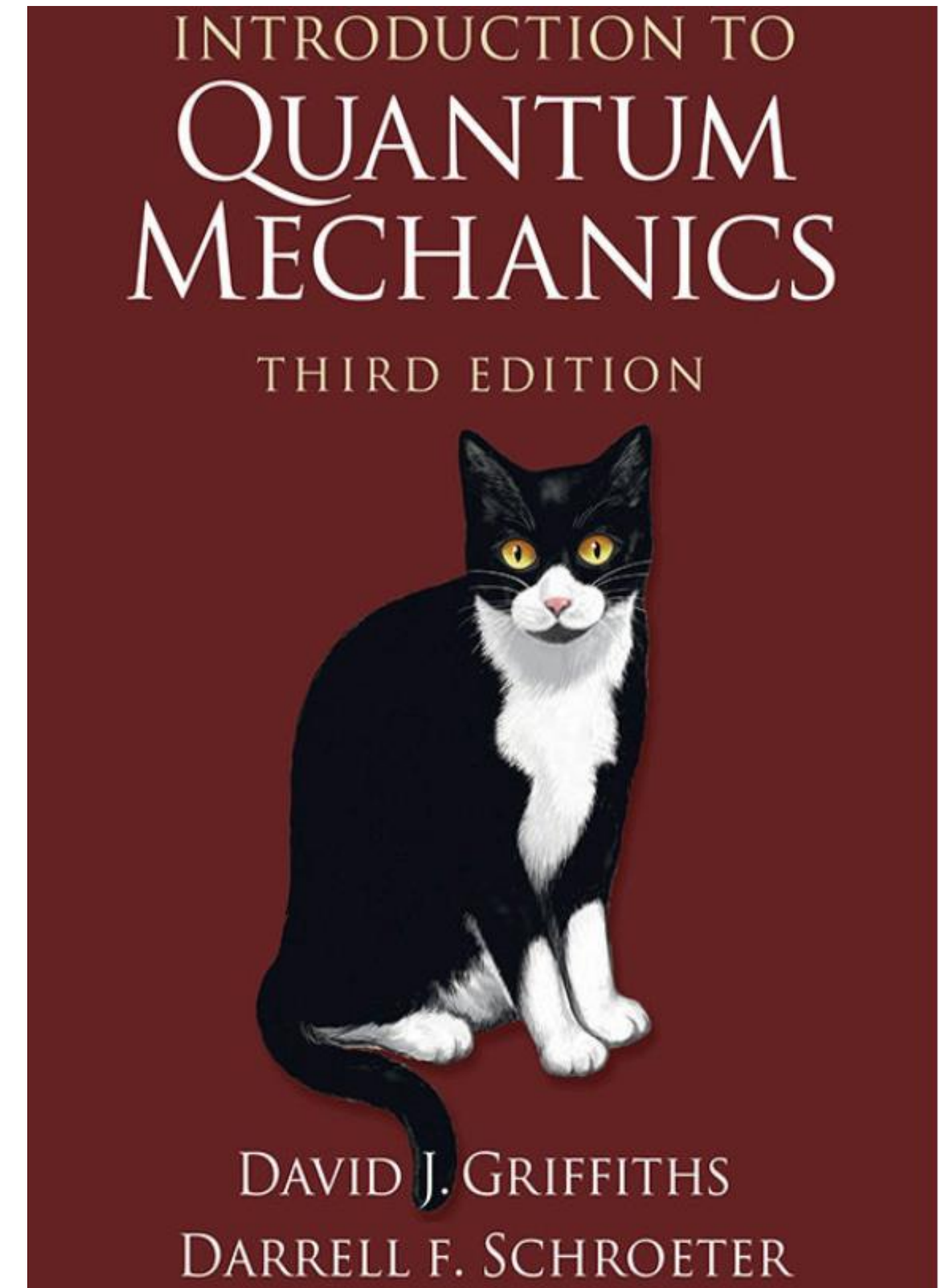
Course book

D.J. Griffiths, Introduction to Quantum Mechanics, Pearson, 3rd edition

Supplementary material

D.A.B. Miller, Quantum Mechanics for Scientists and Engineers, Cambridge

C.C. Gerry and P.L. Knight, Introductory Quantum Optics, Cambridge, 2005



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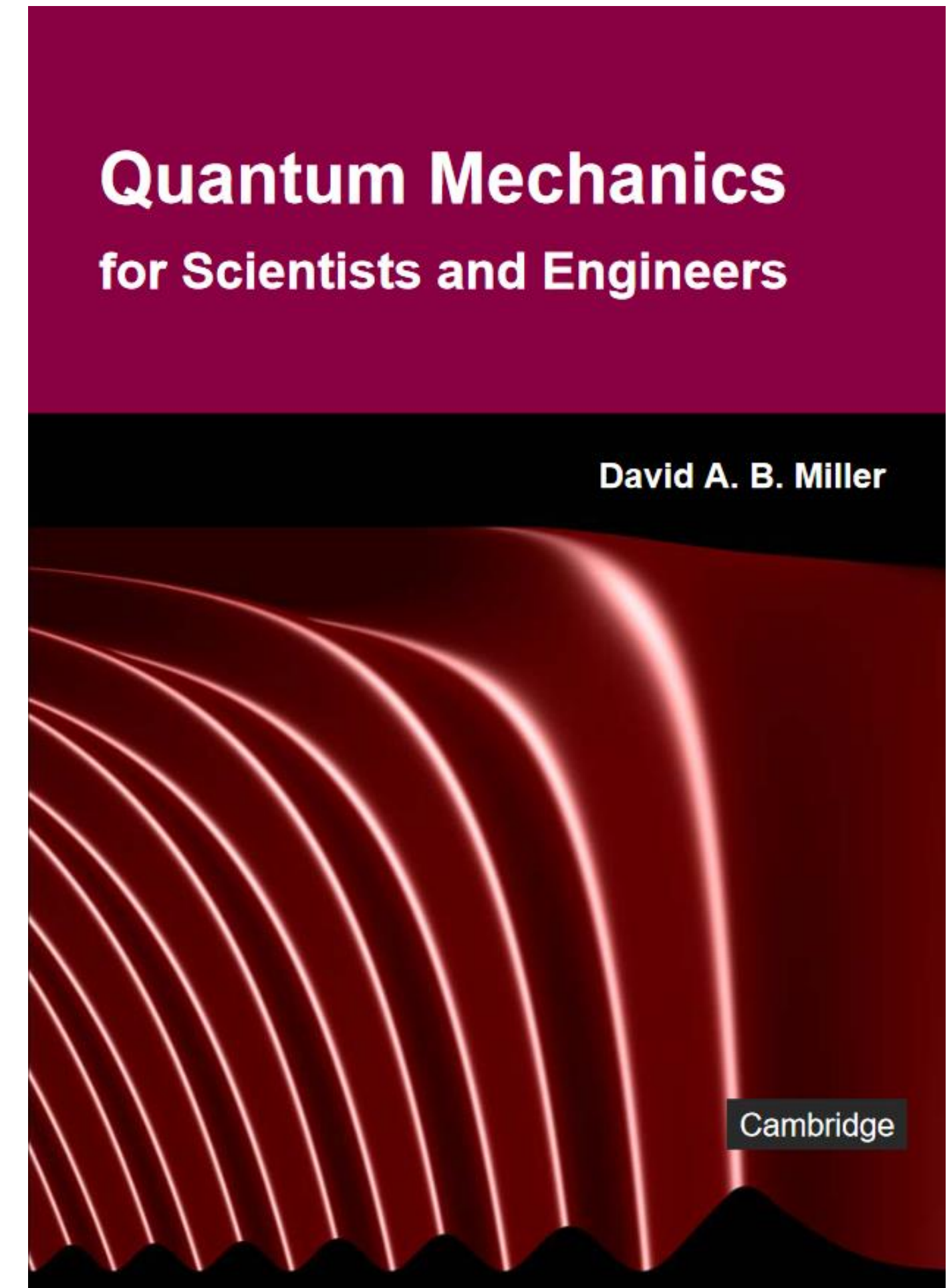
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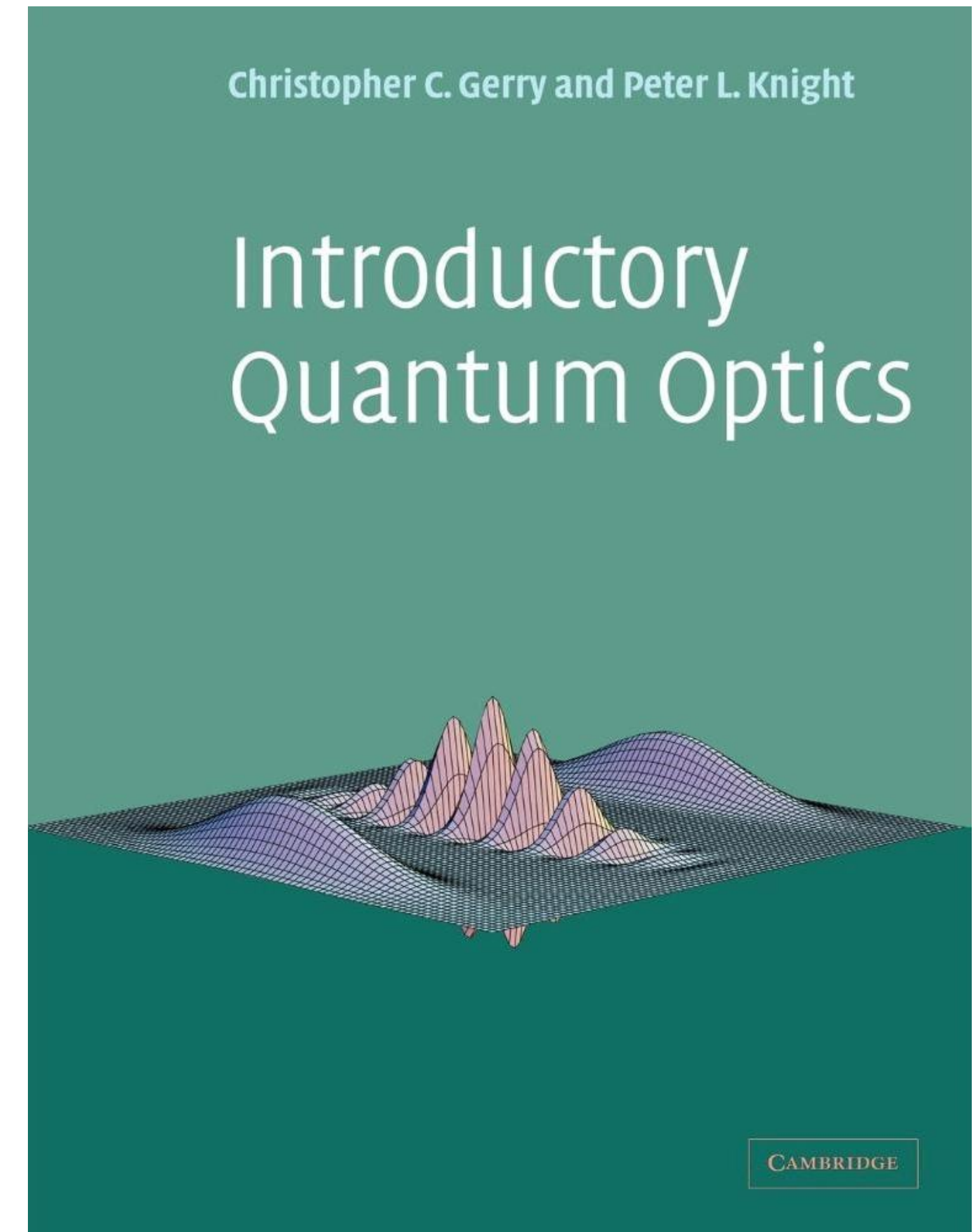
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COURSE MATERIALS

Supplementary material

David Miller's webpage on quantum mechanics

<https://dabm.stanford.edu/teaching/quantum-mechanics/>

QuVis: Web site with quantum mechanics simulations

QuTip: Python library quantum mechanical simulations

Applet(s) by Paul Falstad for [1D quantum systems](#) (other applets available on www.falstad.com/)

OVERVIEW OF THE COURSE

week	Topic	Griffiths
Week 1	Introduction & Required mathematical methods. Waves and Schrödinger's equation	Ch. 1
Week 2	Probability, Uncertainty and Time evolution, Infinite Square well	Ch. 2
Week 3	The Harmonic oscillator, Creation & Annihilation operators	Ch. 2
Week 4	Free particle, 1D Bound states & Scattering/Transmission, Finite well	Ch. 2
Week 5	Approximation methods in 1D	Ch. 3 & 7
Week 6	Quiz 1 + Quantum mechanics formalism: Functions & operators	Ch. 3
Week 7	Dirac formalism, Uncertainty	Ch. 3
Week 8	Angular momentum and Hydrogen atom	Ch. 4
Week 9	Spin	Ch. 4
Week 10	Quiz 2 + Magnetic fields, The Pauli equation, Minimal Coupling, Zeeman Effect	Ch. 4
Week 11	Identical particles	Ch. 5
Week 12	Periodic structures, Band structure, Bloch functions	Ch. 5
Week 13	Time-dependent perturbation	Ch. 7
Week 14	Absorption, spontaneous emission, and stimulated emission	Ch. 7

COURSE SYLLABUS AND CLASS WORKFLOW

Homeworks

- Not graded, idea is to understand the topic better
- You can submit solutions/ask questions, working together allowed

Quizzes (2)

- $2 \times 20\% = 40\%$ of your grade, same style as the final but only 3 questions and you have 1 hour

Final Exam

- 60% of your grade, 7 theoretical open questions, Exercises are important, topics from the whole year

COURSE SYLLABUS AND CLASS WORKFLOW

Quiz dates (tentative)

- Quiz 1: Wednesday 05.11.2025
- Quiz 2: Wednesday 03.12.2025

Final exam (all topics): date TBA